

BOGORODITSKOE-FENINO, Russian Federation

WMO: 341100

Lat: 51.167N

Lon: 37.350E

Elev: 226

StdP: 98.64

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-19.6	-25.3	0.4	-22.1	-22.7	0.5	-19.4	7.9	-6.4	7.3	-5.7	2.1	50	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.0	20.1	30.0	19.4	28.2	18.9	21.3	28.5	20.6	27.5	19.9	26.5	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	24.2	18.3	13.5	23.1	17.7	13.0	22.5	62.9	28.6	60.4	27.5	57.9	26.7	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	5.9	-25.0	34.6	4.0	2.2	-27.9	36.2	-30.2	37.5	-32.4	38.7	-35.3	40.3		
(5)			-25.2	22.8	3.9	0.7	-28.0	23.3	-30.3	23.7	-32.5	24.1	-35.3	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-7.6	-5.5	-0.6	8.7	15.9	19.0	20.8	19.7	14.1	6.9	1.6	-3.1		
	DBStd	11.01	6.43	6.34	4.84	4.66	3.67	3.59	3.28	3.82	3.99	4.66	4.77	5.61		
	HDD10.0	2182	545	434	329	81	3	0	0	0	11	120	254	405		
	HDD18.3	4170	803	668	587	291	91	36	14	28	133	354	503	663		
	CDD10.0	1294	0	0	0	41	187	270	336	302	134	23	1	0		
	CDD18.3	242	0	0	0	1	16	56	91	72	7	0	0	0		
	CDH23.3	2448	0	0	0	13	186	494	824	810	121	0	0	0		
	CDH26.7	774	0	0	0	1	31	121	287	318	17	0	0	0		
	WSAvg	2.8	3.1	3.2	3.2	3.1	2.6	2.4	2.2	2.2	2.5	2.8	3.0	3.3		
	PrecAvg	559	39	30	30	40	49	63	66	60	51	43	48	44		
	PrecMax	792	91	56	71	107	119	138	126	125	111	87	126	110		
	PrecMin	368	4	2	8	4	6	14	13	8	5	1	4	3		
	PrecStd	100	22	13	16	25	27	32	28	33	31	25	30	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.6	14.8	24.8	29.5	31.6	35.0	36.3	28.7	20.8	13.8	8.0		
		MCWB	2.9	3.6	8.6	13.9	17.6	19.8	20.3	19.8	17.3	14.1	11.5	7.0		
	2%	DB	2.5	3.3	9.8	22.3	27.3	29.7	32.0	32.5	26.4	18.5	12.0	6.3		
		MCWB	1.6	2.2	5.8	12.5	16.7	19.3	20.4	20.0	16.3	13.0	10.5	5.6		
	5%	DB	1.4	2.1	7.3	20.0	25.5	28.0	29.8	30.1	24.6	15.7	9.9	4.7		
		MCWB	0.7	1.2	4.3	11.7	16.4	19.3	20.0	19.3	15.6	12.0	8.4	4.0		
	10%	DB	0.4	1.3	5.2	17.2	23.5	26.3	27.7	27.9	22.0	13.7	7.7	3.3		
		MCWB	-0.2	0.5	3.0	10.2	15.7	18.6	19.3	18.6	14.8	11.1	6.7	2.6		
	0.4%	WB	3.3	3.9	8.7	15.2	19.5	21.9	22.5	22.1	19.4	15.3	12.0	7.2		
		MCDB	3.7	4.3	13.3	23.7	25.8	28.0	31.2	30.2	25.4	18.8	13.2	7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.8	2.6	6.2	13.1	18.5	20.9	21.5	20.9	17.3	13.9	10.6	5.7		
		MCDB	2.3	3.1	8.9	20.6	24.7	27.3	28.7	29.7	24.3	17.0	12.0	6.3		
	5%	WB	0.8	1.4	4.5	11.9	17.5	20.2	20.7	20.0	16.2	12.6	8.6	4.2		
		MCDB	1.3	1.8	6.8	18.1	23.3	26.3	27.3	28.5	22.3	15.4	9.7	4.6		
	10%	WB	-0.2	0.6	3.3	10.9	16.5	19.3	20.0	19.2	15.1	11.1	6.9	2.7		
		MCDB	0.4	1.1	5.1	16.7	21.6	24.7	26.2	26.5	21.3	13.4	7.8	3.2		
	5% DB	MDBR	5.0	5.2	6.6	10.2	11.1	10.8	10.7	11.9	10.4	7.6	4.7	4.1		
		MCDBR	3.7	4.1	9.8	13.9	14.1	13.2	14.0	15.3	15.4	11.3	7.0	4.4		
		MCWBR	3.6	3.6	6.3	6.7	6.1	5.6	5.0	5.6	6.9	6.6	5.9	4.3		
		MCWBR	3.5	3.6	8.9	12.5	11.4	11.3	11.6	13.7	13.1	10.1	6.7	4.3		
	5% WB	MDBR	3.4	3.4	6.2	6.5	5.7	5.3	4.9	5.5	6.9	6.6	5.8	4.3		
		MCDBR	3.7	4.1	9.8	13.9	14.1	13.2	14.0	15.3	15.4	11.3	7.0	4.4		
		MCWBR	3.6	3.6	6.3	6.7	6.1	5.6	5.0	5.6	6.9	6.6	5.9	4.3		
		MCWBR	3.5	3.6	8.9	12.5	11.4	11.3	11.6	13.7	13.1	10.1	6.7	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.338	0.367	0.446	0.401	0.390	0.424	0.430	0.392	0.361	0.338	0.308		
	taud		2.124	2.041	2.104	2.071	2.291	2.341	2.257	2.224	2.303	2.348	2.281	2.186		
	Ebn at Noon		678	754	815	789	854	866	827	798	788	741	647	613		
	Edh at Noon		77	111	126	148	124	119	128	126	104	83	68	63		
(44) All-Sky Solar Radiation	RadAvg		0.95	1.95	3.25	4.16	5.71	6.04	5.81	5.12	3.38	1.82	0.80	0.58		
	RadStd		0.20	0.29	0.52	0.39	0.44	0.50	0.41	0.44	0.43	0.28	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.82	N/A	N/A	N/A	N/A	N/A	-157	-226	+122	+50

Nomenclature: See separate page